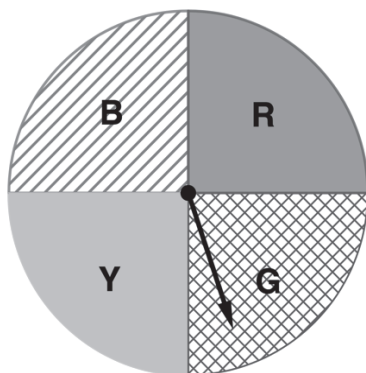


Grade 7
Unit 2
Lesson 7 Practice

Name _____
Date _____
Period _____

1. Explain the difference between theoretical and experimental probability of a simple experiment.

Jennifer performed a simulation and spun the arrow on the spinner below. She recorded her results in the frequency table below.



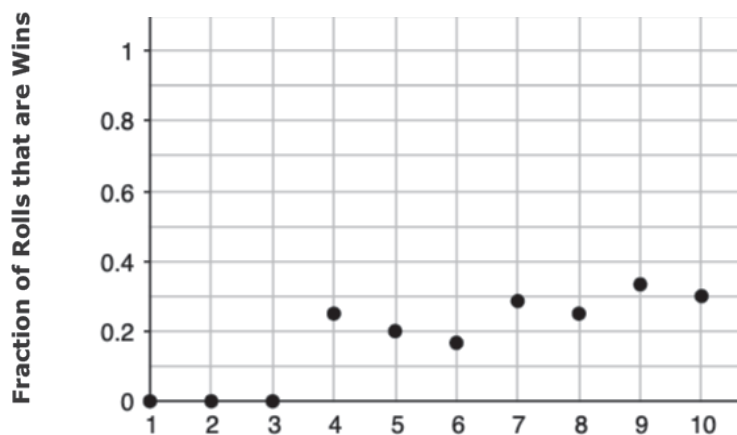
Color	Frequency
Blue (B)	8
Green (G)	3
Red (R)	7
Yellow (Y)	4

2. What is the theoretical probability of landing on green?
3. What is the experimental probability of landing on green?

Tyler is playing a game where he wins if he rolls a 2 or a 5 on a standard number cube.

4. What is the theoretical probability that Tyler wins the game?

He then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.
6. What type of probability is this?
7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 2 or a 5.
8. After 10 rolls, what fraction of the total rolls were a win?
9. If Tyler rolled the number cube 10 more times, estimate the probability that he will roll a 2 or a 5.
10. If he were to roll the number cube 500, how many times would you expect him to roll a 2 or a 5.